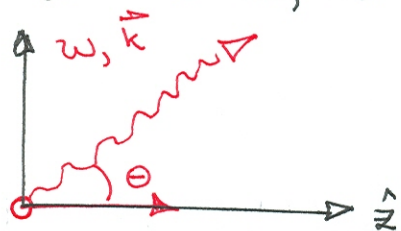


1.11

$\chi^\mu = (\omega/c, \vec{k})$ is a 4-vector, derive the dopler shift



Source moving
 $\vec{v} = \beta c \hat{z}$

$$\chi^{\mu'} = \Lambda^\mu_{\nu} \chi^\nu$$

transform back to rest frame

$$\omega_0/c = \gamma \omega/c - \gamma \beta k_z$$

$$\omega_0 = \gamma \omega - \gamma \beta k_z c$$

$$\omega_0 = \gamma \omega - \gamma \beta \omega \frac{k_z}{k}$$

$$\omega_0 = \gamma \omega (1 - \beta \cos \theta)$$

$$\omega = kc$$

$$\frac{k_z}{k} = \cos \theta$$

$$\boxed{\frac{\omega}{\omega_0} = \frac{1}{\gamma(1 - \beta \cos \theta)}}$$